Explicit sporadic markings and their incidence certificates

Human Compression project

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Abstract

We give reproducible marked permutation groups for the sporadic application in Theorem 5.3 of Piedade–Tranchida, in the edition arXiv:2605.04646v2. The finite checks distinguish group identification, the intersection property, and flag transitivity. An obstruction to the particular O'Nan duality route in that proof is treated separately from the existence assertion.

1 Scope and retained inputs

The source theorem asserts that, for every sporadic simple group H admitting a proper almost-simple extension, its full automorphism group admits both a regular rank-three polytope and a regular rank-three hypertope with a nonlinear diagram, except possibly for Fi_{22} and Fi'_{24}. The nine cases explicitly treated in the source proof are

$$M_{12}, J_2, J_3, HS, He, Suz, \mathrm{O'N}, M_{22}, McL.$$

The literal theorem also includes the Harada–Norton socle HN, which has an outer automorphism of order two but receives no case in the source proof. It remains unpaid by this nine-case packet. We preserve the two stated Fischer exceptions separately. There is no assertion of new priority for the existence theorem, nor of a resolution of either exception.

The naming of the groups uses the GAP primitive permutation catalogue and the Atlas standard permutation representations. The classification of sporadic outer automorphism groups, including $|\operatorname{Out}(H)| = 2$ in these nine cases, remains a retained finite-group input. The GAP permutation algorithms are used as computational proof infrastructure, not as Lean proofs. The identification of an arbitrary concrete permutation group from its order alone would be insufficient; the checks below additionally bind the named representation or the stated presentation. No attempt is made to reprove the character table of $\mathrm{O'N} : 2$, the Atlas classification, or the polytope–string-C-group correspondence.

All new witness permutations and words, and the exact checks upon them, are explicit. The earlier large atlas of polytopes is not a proof premise. In particular, a random search is merely a way of finding a candidate; the deterministic verification of the saved candidate is the certificate.

2 The incidence checks

Let $r = (r_0, r_1, r_2)$ consist of distinct nonidentity involutions generating a finite group G. Put $G_I = \langle r_i : i \in I \rangle$ and $P_i = G_{\{0,1,2\}\setminus\{i\}}$. We check

$$G_I \cap G_J = G_{I\cap J} \qquad (I, J \subseteq \{0,1,2\}), \tag{1}$$

including all 64 ordered pairs of subsets. A pairwise intersection of the three maximal parabolics is not accepted as an unexplained substitute for the full condition.

For every permutation (i, j, k) of $(0, 1, 2)$ with fixed first entry, we also check the equality of actual finite sets

$$(P_iP_j) \cap (P_iP_k) = P_i(P_j \cap P_k). \tag{2}$$

Only three equalities are necessary, since the last two indices are interchangeable. These are product sets, not generated subgroups.

Here is the direct flag interpretation. A flag containing the base i-element is described by intersecting cosets of the other two parabolics. Translating the incidence with the base element reduces the remaining common-representative question to (2). Consequently every three-element flag has a common representative; two-element flags have one by the definition of incidence. Thus the coset system is flag-transitive. Condition (1) identifies every standard residue stabilizer and makes each panel have two chambers. In a rank-two residue, its two generating involutions generate the corresponding dihedral group; its coset incidence graph is therefore connected. The whole incidence graph is connected because the marking generates G. These statements establish thinness and residual connectedness, in addition to flag transitivity. The trivial intersection of the maximal parabolics makes the action on chambers faithful. The group is chamber-regular and is the full type-preserving automorphism group: an automorphism fixing one chamber fixes its uniquely determined adjacent chambers and then every chamber by connectedness.

If $r_0r_2 = r_2r_0$, the marking is string and hence gives the regular polytope. For the second marking every pair product has order greater than two; its rank-three diagram is a triangle.

3 The six outer-dual rank-four cases

The saved tuple $r = (a, b, c, d)$ generates the indicated simple group H. We verify all 256 ordered intersection equalities, the string commutations, and a saved involution t satisfying

$$a^t = d, \quad b^t = c, \quad c^t = b, \quad d^t = a.$$

The checks give $t \notin H$, $G = \langle H, t\rangle$, and $C_G(H) = 1$. Thus conjugation embeds G into $\mathrm{Aut}(H)$ and the retained outer-order fact identifies it with the full automorphism group. In particular conjugation by t is not inner on H: otherwise th^{-1} would centralize H for some $h \in H$.

Both rank-three markings are checked directly:

$$(a, t, c) \qquad \text{and} \qquad (a, b, t).$$

This avoids making their validity depend on a convention for orbit representatives in the source's folding construction.

H	degree	$\lvert H\rvert$	rank-four type	string orders	triangle orders
M_{12}	144	95040	$\{6, 3, 6\}$	$(4, 2, 6)$	$(6, 4, 6)$
J_2	100	604800	$\{6, 3, 6\}$	$(4, 2, 6)$	$(6, 4, 6)$
HS	100	44352000	$\{8, 3, 8\}$	$(4, 2, 6)$	$(8, 4, 6)$
He	2058	4030387200	$\{6, 6, 6\}$	$(4, 2, 12)$	$(6, 4, 12)$
Suz	1782	448345497600	$\{6, 5, 6\}$	$(4, 2, 10)$	$(6, 4, 10)$
J_3	6156	50232960	$\{5, 9, 5\}$	$(4, 2, 18)$	$(5, 4, 18)$

The order triples list the products in positions $(01, 02, 12)$. The first five named full extensions are respectively `PrimitiveGroup(144,4)`, `PrimitiveGroup(100,2)`, `PrimitiveGroup(100,4)`, `PrimitiveGroup(2058,2)`, and `PrimitiveGroup(1782,2)`. The $J_3 : 2$ witness uses the Atlas standard generators of degree 6156; all eight relations of the Atlas presentation and the exact group order are checked. The retained standard representation is bound separately from the new marking.

4 The Mathieu and McLaughlin cases

For $M_{22} : 2$ we use exactly the three degree-22 permutations printed in the source. The resulting string marking has product orders $(4, 2, 12)$ and generates order 887040. Its derived subgroup is simple of order 443520 and is explicitly isomorphic to `MathieuGroup(22)`. The halved triple

$$(r_0, r_1, r_1^{r_2})$$

has product orders $(4, 4, 6)$, generates the same group, and passes (1)–(2).

For $McL : 2 =$ `PrimitiveGroup(275,2)` the saved triple (t, a, b) has product orders $(6, 2, 5)$. Its halved triple (t, a, a^b) has orders $(6, 6, 5)$. Both generate the full group of order 1796256000 and pass the same complete tests. The derived subgroup is simple of order 898128000, and its centralizer in the full group is trivial. The search uses an outer involution and involutions in its centralizer, as suggested by the source, but the final saved triple no longer requires a search.

For completeness, if a, b are involutions with $|ab| = k$ odd, then

$$b = (abab)^{(k-1)/2}a.$$

Since $abab = a(a^b)$, the halved pair a, a^b recovers b. This proves generation preservation in the odd-label case; it does not by itself prove the intersection property or flag transitivity, which are independently checked.

5 The O'Nan proof-route obstruction

Lemma 1. *Let H have an index-two extension G with no order-four element outside H. A string marking of H of rank at least three cannot admit an involutory duality induced by conjugation by an element $t \in G \setminus H$.*

Proof. Write the endpoints as a, c. They are distinct commuting involutions. If $a^t = c$ and $t^2 = 1$, then $(at)^2 = ac$ is a nonidentity involution. Hence at has order four and lies outside H, a contradiction. □

For $G = \mathrm{O'N} : 2$, the nontrivial degree-one character of the retained Atlas character table has value -1 on precisely the outer coset. The orders of its classes there are

$$2, 6, 8, 10, 14, 22, 24, 30, 38, 56.$$

The checker derives this list from the named table, verifies that the degree-one character is unique and has values in $\{1, -1\}$, and records the table's provenance. It follows that the rank-three outer-duality route stated on source page 19 cannot have the asserted marking. This is an obstruction to that proof route, not a disproof of the source's existence theorem.

A direct construction

Let t, d be the hash-bound Atlas standard generators of O′N : 2 on 245520 points, of orders $2, 4$, with $|td| = 22$. The file `ON_compact.g` specifies eight short words w_j, one word w_a, ten indices

$$(6, 6, 5, 7, 5, 4, 2, 4, 5, 4),$$

and the exponent 5. Signed letters $1, 2$ mean t, d, with negative letters denoting inverses. Put $h_j = w_j(t, d)$, $x_j = tt^{h_j}$, and $n_j = |x_j|$. Every n_j is odd. Define

$$z_j = h_j x_j^{(n_j-1)/2}, \qquad a = w_a(t, d), \qquad b = (z_6 z_6 z_5 z_7 z_5 z_4 z_2 z_4 z_5 z_4)^5.$$

The checker verifies every commutation $[t, z_j] = 1$ directly. This is Bray's odd-commutator centralizer construction[2], used here only to specify a short deterministic witness. Its containment identity is elementary: if $x = tt^h$ has odd order n and $k = (n-1)/2$, then $t^h = tx$, $txt = x^{-1}$, and $t^{hx^k} = x^{-k}tx^{k+1} = tx^{2k+1} = t$. No assertion that arbitrary chosen words generate the full centralizer is needed by the certificate.

The resulting marking (t, a, b) has orders $(8, 2, 11)$ and generates the full group of order 921631011840. Its saved construction uses words in the Atlas generators, and the reverse membership of d is checked; equality is therefore with the actual named group, not merely a group of the same order. Both the intersection property and flag transitivity hold. The halved marking (t, a, a^b) has orders $(8, 8, 11)$. Its generation follows by the checked word

$$b = \big(a(a^b)\big)^5 a,$$

and its full intersection and flag-product equalities are again checked directly. This establishes the positive O'Nan conclusion without the impossible outer-duality premise. No unsuccessful search is used as evidence of nonexistence.

6 Verification boundary

The verification scripts use GAP's exact finite permutation operations. All nine cases have deterministic saved witnesses. The O'Nan proof-route obstruction is a separate check. Programs that search for witnesses are historical construction aids and are not required for replaying a fixed witness. The entire finite checker is retained, including generation and every subgroup intersection; it is not a sample test. The external named-group and character-table inputs remain visible. The external ON representation pair has 3,720,665 bytes, in addition to the 1,948-byte word instructions. Both inputs and the installed catalogue files are hash-bound in the retained-input ledger; their classification proofs are not priced as zero. There are no Lean declarations in this finite packet, and no assertion that GAP calculations have been translated into a theorem-prover kernel.

References

[1] C. A. Piedade and P. Tranchida, *The geometry of wreath and semi-direct products*, arXiv:2605.04646v2, 28 May 2026.

[2] J. N. Bray, An improved method for generating the centralizer of an involution, *Arch. Math.* **74** (2000), 241–245, doi:10.1007/s000130050437.

[3] J. H. Conway, R. T. Curtis, S. P. Norton, R. A. Parker and R. A. Wilson, *Atlas of Finite Groups*, Oxford University Press, 1985.

[4] R. A. Wilson and collaborators, *Atlas of Finite Group Representations*, `https://brauer.maths.qmul.ac.uk/Atlas/`; exact per-file URLs are supplied in the retained-input ledger.

[5] The GAP Group, *GAP—Groups, Algorithms, and Programming*, version 4.15.1, with PrimGrp 4.0.1 and CTblLib 1.3.11. These exact versions and data files are bound in the ledger.